Learning Safe Adapters Around Frozen World Models

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ABSTRACT

Can evolutionary search learn compact adapters that recover safe task completion around frozen world-model systems, reducing reliance on manually tuned behavioural rules? We test recurrent action adapters around LeWorldModel planning and a candidate selector around frozen UNISafe Block Plucking. On TwoRoom, CMA-ES, random search and PPO achieved 44, 37 and 20 safe completions in 216 evaluation episodes each. All received the same adapter information and 40,000 training/selection actions per repeat. The exploratory contrast intervals include zero, and a separate mechanism study slightly favoured random search. PushT baseline reproduction recovered 88/128 completions with the released evaluator and 89/128 with our wrapper; subsequent hazard qualification passed, but adapter authority failed. The Block Plucking pilot found no pooled gain over native. A TwoRoom policy library offered some risk tradeoffs, but five of 28 available settings exceeded the requested empirical risk limit. Evolutionary superiority, reduced tuning, calibrated risk and reliable cross-environment gains remain unestablished. Further development studies are active.

KEYWORDS

evolution strategies, safe task completion, world models, adapters, reinforcement learning

1 RESEARCH QUESTION AND SCOPE

A policy can avoid violations by failing to finish. Conversely, a controller can finish often while violating a safety requirement. Our primary endpoint is therefore *safe task completion*, accompanied by completion, violations, unfinished episodes and engagement. We ask whether a compact learned adapter can improve this endpoint around a frozen world-model system with less manual behavioural design. Matching a carefully tuned reference with demonstrably less tuning would be useful; beating every specialised controller is not a necessary condition.

Useful *authority* requires parameter changes to alter safe completion. Learning must improve over native and untrained references; an evolutionary advantage additionally requires comparison with parameter random search and ordinary learning under matched information and interaction budgets. Reduced tuning requires a separate accounting of development effort and tuning evaluations.

LeWM provides the opening experiment, starting with TwoRoom and extending to PushT when baseline and authority checks permit. Block Plucking is the intended main experiment because violations have physical task definitions. Cube is deferred after failed task-completion and renderer checks. ManiGuard and the October planner studies are historical context, not additional repeats of the present experiment. The connection between LeWM and Block Plucking is adaptation around a frozen world model. Their currently implemented adapters differ, so we do not claim shared weights or an identical architecture.

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2 RELATION TO PRIOR WORK

LeWorldModel learns a latent predictive model from pixels [1]; our work freezes the released backbone and adapts its use. Anchored Planning retrieves short observation targets from offline trajectories to help frozen planners reach long-range goals [2]. It supplies our baseline planning mechanism, not our learning contribution. UNISafe combines uncertainty detection and latent reachability to filter unsafe actions [3]. Our Block Plucking study uses its frozen components and physical task, with filter behaviour documented separately. We use established evolutionary and policy-gradient optimizers, CMA-ES [4] and PPO [5]. The research question concerns their utility for compact safety adaptation under matched information and interactions, not novelty of either optimizer.

Separating task reward from constraints is established in constrained RL, including Constrained Policy Optimization [6]. A requested risk budget is also an established formulation: budgeted RL allows its constraint budget to vary at execution time [7]. Our current PPO arm is a matched-utility optimizer control, not a comparison against these constrained methods. Likewise, selecting an existing finite library by empirical violations is not the same as learning a budget-conditioned value function or inheriting a training-safety guarantee. Evolutionary constraint handling also predates this work. Deb's feasibility rules prefer feasible candidates, compare feasible candidates by objective, and compare infeasible candidates by violation, avoiding a penalty coefficient [8]. Stochastic ranking instead uses a probability to balance objective-based and violation-based comparisons [9]. Removing a penalty coefficient therefore does not by itself establish novelty or remove every design choice. Our current fixed utility implements neither of these constraint-handling methods. The intended contribution is measured adaptation and tuning cost around frozen models.

Evolution also has direct safe-RL precedents. Safe Neural Evolutionary Strategies use Bayesian model checking to adapt a Lagrangian tradeoff according to confidence in constraint satisfaction [10]. Safety-Oriented Search augments deep RL with evolutionary cost optimization and mutations informed by visited unsafe states [11]. Our experiments instead compare optimizers of compact adapters around frozen world-model systems. They do not implement these two approaches, compare against them, or inherit their verification procedures. Evolution for safety and adaptive tradeoffs alone therefore cannot constitute our novelty claim.

Multi-objective methods address a broader tradeoff-learning problem. MO-ERL combines policy-gradient learning with evolutionary search to maintain diversity across a Pareto front [12]. Constraint-Rectified Multi-Objective Policy Optimization alternates multi-objective policy improvement with updates that reduce an

unsatisfied constraint; its convergence analysis is tabular [13]. Our library instead retains fixed-utility search candidates and selects among them afterward. It neither learns a Pareto front nor implements these methods. Probabilistic safe RL also distinguishes trajectory-level safety from cumulative-cost relaxations [14]. Our episode indicator targets any violation, but finite development counts do not certify its population probability.

3 PROBLEM AND CONTROLS

For LeWM episode i, let C_i indicate task completion and V_i indicate any monitored violation, including a violation before eventual completion. Safe completion is $S_i = C_i(1-V_i)$. We report all denominators. Violations may overlap completed and unfinished episodes; these quantities are not an additive partition. Movement is reported as engagement and is not treated as successful progress.

The main LeWM monitor labels executed trajectories and never vetoes an action. Hazard geometry is supplied equally to the relevant learning arms. Safe scripted witnesses establish feasibility and risky witnesses establish an avoidable violation; they are privileged diagnostics, not demonstration inputs to the adapter. A useful case must support a safe route and give the baseline an opportunity to take risk.

All weights of the backbone, retrieval memory and planner are frozen. Changes to query panels, memory exclusions, precision, renderer and horizon can change baseline competence. We therefore qualify them before searching for a safety controller. Protocols, seeds, panels, budgets and gates are fixed before their outcomes. Every simulator action counts, including selection, discarded candidates and failed starts. Model queries and elapsed time are recorded separately. Comparisons are blocked by host when numerical behaviour differs across hardware.

4 OPENING STUDY: LEWM

4.1 Adapter and learning

The TwoRoom adapter encodes current and goal latents, normalized geometry, the base action, explicit lag inputs and time. A 32-unit feature layer and 32-unit GRU produce an action gain and residual. For base action a_t^0,

$$a_t = \mathrm{clip}_{[-1,1]}\left[(1+\tanh g_t)\odot a_t^0 + \tanh r_t\right]. \tag{1}$$

Zero output heads recover the base controller exactly. The actor has 19,172 parameters in TwoRoom; a development comparison selected its 132 head coordinates for the subsequent matched studies. Frozen random features and recurrence still contain capacity, so a random untrained adapter is an essential reference. PushT uses 402 inputs instead of 396, with its task geometry normalized explicitly. This is reuse of an interface family, not same-weight transfer.

Diagonal CMA-ES uses population eight, initial standard deviation 0.5 and explicit cumulative step-size adaptation. Random search samples the same initial distribution. On complete eight-case panels, $81\sum_i S_i + 9\sum_i C_i - \sum_i V_i$ orders safe completion, completion and fewer violations lexicographically: $9 > 8$ and $81 > 9\cdot 8 + 8$. The coefficients encode this chosen hierarchy; they were not tuned. Partial panels consume budget but cannot nominate a policy.

Table 1: Matched-objective TwoRoom study. Native is shared within each host; the other rows pool three learner repeats on the same 24 layouts and three planner seeds. Episodes are not independent task samples. All episodes engaged.

Controller	Safe	Complete	Violated	Unfinished	n
Native	14	77	75	67	144
Untrained	34	127	111	89	216
CMA-ES	44	116	109	100	216
Random	37	99	84	117	216
PPO	20	110	116	106	216

The PPO comparison trains the same head-coordinate actor with a separate critic and stochastic raw logits; evaluation exports a deterministic actor. Its terminal reward is $(81S_i + 9C_i - V_i)/90$, without a dense progress reward. Training and selection together receive exactly 40,000 actions per optimizer per repeat. PPO also learns four exploration-scale parameters and the critic; the comparison matches actor coordinates, information and interactions, not total training parameters. The finite-panel ordering does not imply lexicographic population probabilities: PPO maximizes expected fixed utility. Safety specification and preference design remain, and reduced tuning is untested. Only one PPO configuration was tested.

4.2 TwoRoom evidence

The initial replication found CMA safe completion of 8/36, 6/36 and 11/36 across three repeats, compared with native 3/36 and random search 4/36, 4/36 and 1/36. A later representation/mechanism study selected head coordinates and evaluated on 24 new layouts with three planner seeds. It found 45/216 safe completions for CMA and 47/216 for random search, against shared native 7/72. One untrained reference outperformed its paired evolved adapter. These outcomes limit any conclusion drawn from the initial positive pilot.

The separate matched PPO study used two complete optimizer comparisons on one RTX 4090 and one on an L4. CMA ranked highest in aggregate (Table 1) and exceeded PPO in every repeat. A pre-specified crossed bootstrap resampling learner repeats and layouts, with planner seeds retained together, gave CMA minus PPO +11.11 percentage points, exploratory 95% interval $[-6.94, +28.24]$; CMA minus random was +3.24 points, interval $[-11.57, +18.52]$. Three repeats and shared final layouts limit inference. This supports neither broad evolutionary superiority nor a claim that PPO cannot learn the task.

The mechanism study also examined task-only ranking, removal of online hazard geometry and recurrent-state resets. Their safe-completion totals were 29/216, 36/216 and 37/216, respectively, with mixed effects across repeats. Hazard relocation changed 72/72 main-adapter action sequences and 0/72 hazard-blind sequences. This establishes response to hazard information, not necessarily a safer response.

4.3 PushT baseline and qualification

A first distinct-scene screen excluded entire scene groups from an 8,000-episode retrieval memory. Native completed 4/16 episodes,

safely in 2/16, with 4/16 violations and 12/16 unfinished. The prerequisite gate failed, so no parameter probes or learning followed. That result is retained.

We then reproduced the released AP-CEM protocol on its 128 queries with its full 16,688-episode memory, published package versions and fixed seeds. The released evaluator completed 88/128 and the existing wrapper 89/128; both engaged in all 128 episodes, with no initially successful queries. They shared 88 successes and 39 failures. Query episodes were excluded, but sibling demonstrations from the same scene could remain. The study used RTX 4090 rather than H20 hardware and a different Python patch release. Near-matching aggregate performance does not imply bitwise equivalence.

The earlier screen and reproduction differ in memory, queries and runtime together. Their difference does not isolate a causal explanation. The reproduced baseline supports a new episode-excluded, familiar-scene qualification; it does not validate novel-scene generalization. That qualification fixes eight development scenes and twelve reserved candidate final scenes, with safe and risky witnesses and sixteen pre-drawn head-coordinate probes. The completed 329-episode qualification passed its independent audit. Native completed 12/16 episodes but violated in all 16; five scenes passed admission. Across sixteen head-coordinate probes, safe completion on the admitted subset ranged only from 0/10 to 1/10 and occurred on just one scene. Across all scenes, probe completion ranged from 0/16 to 4/16, below native 12/16, despite engagement in every episode. The useful-authority gate failed, so no matched optimizer training followed and the twelve reserved cases remain untouched. This separates baseline competence from useful control authority in the tested adapter space.

5 INTENDED MAIN STUDY: BLOCK PLUCKING

Block Plucking uses frozen UNISafe components and task-native physical violations. Here "native" denotes the bare anchored planner. The safety filter computes shadow signals but never vetoes the planner's commands. This is a study built on UNISafe components, not a reproduction of its released filtered-policy evaluation. The primary endpoint also differs from LeWM: after initial benchmark success, the physical success predicate must hold for all 40 additional controls (two seconds), with no violation. Benchmark success, sustained safe success and any violation during continuation are recorded separately. "Unfinished" here means no initial benchmark success, including episodes that ended on a violation. The current pilot selects among retrieved trajectory candidates around a frozen world-model planner, rather than applying the LeWM recurrent action residual. Its five linear weights modify candidate ranking using standardized retrieval distance, imagined cost, mean arm magnitude, closing fraction and demonstration progress. Neutral weights recover the bare planner. This implementation difference is material to the scope of transfer.

The completed three-repeat pilot evaluated 48 fresh starts per controller. Sustained safe success was 10/48 native, 12/48 untrained, 7/48 random search and 10/48 CMA. Violations were 28/48, 25/48, 30/48 and 26/48, respectively. CMA's safe-completion differences from native were +2, +2, −4 across repeats. The pooled pre-registered verdict was no signal. The pilot matched training episode counts, whereas LeWM matches actual simulator actions; these are not identical budgets. CMA and random search also used different training-panel allocation rules in this pilot, which limits optimizer interpretation.

The next Block Plucking study independently changes training-panel breadth, nominee selection, elite re-evaluation and frozen-system candidate scores. Its selector expands from five to eight weights, adding imagined terminal minimum critic value, predicted failure probability and ensemble disagreement along the imagined action block. These are signals from the pretrained frozen system. Its first completed repeat (R1) passed an independent saved-record audit. On 16 final starts, sustained safe completion was 2 native, 6 untrained, 6 random and 5 CMA; violations were 9, 9, 6 and 10, respectively. Benchmark successes were 10, 10, 9 and 7, leaving 6, 6, 7 and 9 unfinished. CMA therefore received the frozen mixed verdict: three more sustained completions than native, but one more violation. Its final mean beat its last elite on the selection panel. Two repeats remain active, so no pooled conclusion follows. Since the changes are bundled, a gain would not identify its cause. Comparisons stay within host because equal seeds produce different trajectories across hosts.

The R1 audit checks all 1,156 completed episode records, including four host-local smoke episodes (two more than the original budget table), search rankings, nominees and final panels. It reconstructs outcomes from saved predicates without replaying the simulator. This run used 93,457 controls. Setup and discarded launches are accounted separately; incomplete attempts provide only recorded lower bounds. Three withdrawn-repeat smoke episodes are documented, but their raw resource records have not been located. Complete study-wide resource use is not verified.

6 SAFETY DIAL AND RELIABLE GAINS

A risk limit is useful only when candidate policies offer different task/risk choices. A zero-rollout diagnostic used complete training panels from the matched PPO study, choosing maximum completion subject to empirical violation limits $0, 0.25, 0.5, 0.75, 1$. It found distinct selected CMA and random policies in all three repeats, but only one selected PPO policy per repeat. Missing targets remained unavailable. These are eight reused selection cases, so their risk estimates cannot certify deployment probabilities.

The prospective test locked 17 distinct selected actors, shared native and three untrained references before 756 episodes on twelve previously observed development layouts and three planner seeds. All 772 records, including 16 engineering episodes, passed audit. Of 45 target assignments, 17 were unavailable; five of 28 available assignments exceeded the requested empirical risk limit. Four of six curves selecting multiple policies had nondecreasing violations; R0 CMA and R2 random search were nonmonotonic. PPO selected one actor per repeat at limits 0.75 and 1, with no lower targets available. Such constant curves do not establish a useful dial.

For example, R0 CMA violated in 18/36 episodes at limit 0.25 and 16/36 at 0.5. R1 CMA offered a risk change from 4/36 to 11/36, but completion was only 11/36 and 12/36, leaving 25/36 and 24/36

unfinished. Every policy engaged in all 36 episodes. Repeated targets reuse the same actor evaluation, so assignments are not independent trials. This is limited development evidence of policy choice, not a calibrated-risk result. The library was trained with the original fixed utility, not these risk limits. Its 360,000 acquisition actions remain charged separately from the new 105,532 actions. No reduced-tuning claim follows.

A separate L4 development study tests population four against the existing eight and matched random search in two repeats, each at 40,000 actions per arm. The hypothesis is that more CMA updates can help at a small interaction budget. Its advancement criterion requires gains over both controls in both repeats without higher violations than population eight. No result is yet available. Passing that criterion would motivate fresh confirmation, not establish a reliable evolutionary advantage by itself.

7 LIMITATIONS AND REMAINING EVIDENCE

The present monitors and inputs still require designed safety definitions. TwoRoom hazards are constructed; PushT protects only the pusher; Block Plucking uses different physical predicates and sustained completion. These settings support mechanistic comparisons, not a universal human-values alignment claim. Familiar scene retrieval and previously observed development layouts restrict transfer claims. Small numbers of learner repeats, model-based numerical variability and one ordinary-learning configuration restrict optimizer conclusions.

Reduced tuning remains a distinct experiment: compare a frozen adaptive method against a specified manually tuned reference, count every development interaction for each, and evaluate on untouched tasks under a predeclared practical comparison criterion. Current fixed utility coefficients and post-diagnostic design choices must remain disclosed. No such benefit is claimed here.

Active development and Block Plucking outcomes remain necessary for final claims. Failed prerequisites stay in the evidence. Current results support useful TwoRoom adaptation and baseline qualification; cross-environment gains, reduced tuning and reliable evolutionary superiority remain unestablished.

EVIDENCE MAP AND DRAFT STATUS

Rule43 accounts 9 and 12–17 index completed LeWM evidence; `SafeBlock/PilotPooled.md` records the Block pilot. The active studies are `Search2.md` and `EvolutionBudgetStudy.md`. Final-result integration, broader literature coverage and author/supervisor review remain before submission.

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